THE DIAGONAL OF THE PICARD FUNCTOR IS REPRESENTABLE BY IMMERSIONS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Proposition proposition-pic-functor (canonical tag 0D2C). cited at statement line 3275. Assumptions and notation: the Picard functor is an fppf sheaf.
- Lemma more-morphisms-lemma-separated-locally-quasi-finite-morphisms-fppf-descend (canonical tag 02W8). cited at proof lines 3299-3301. Descent of the locally quasi-finite Z_i -> T_i to Z -> T.
- Lemma descent-lemma-descending-fppf-property-immersion (canonical tag 02YM). cited at proof line 3304. Z -> T is an immersion.
- Lemma lemma-flat-geometrically-connected-fibres (canonical tag 0D25). cited at proof line 3316. Characterizes Z: factoring through Z iff L_{X_{T'}} is a pullback from T'.
- Lemma spaces-more-morphisms-lemma-diagonal-picard-flat-proper (canonical tag 0D23). cited at proof lines 3317-3319. Existence of the locally closed Z in the invertible-module case.

Source reference. Original proposition 0D2C: sources/stacks-chow/tags.txt, source lines 16377–16377. Cited at statement line 3275: assumptions and notation inherited from this proposition; load-bearing.

Source reference. Original lemma 02W8: sources/stacks-chow/tags.txt, source lines 3795–3795. Cited at proof lines 3299-3301: Z -> T obtained by descending the locally quasi-finite Z_i -> T_i; load-bearing.

Source reference. Original lemma 02YM: sources/stacks-chow/tags.txt, source lines 3881–3881. Cited at proof line 3304: Z -> T is an immersion; load-bearing.

Source reference. Original lemma 0D25: sources/stacks-chow/tags.txt, source lines 16370–16370. Cited at proof line 3316: T' -> T factors through Z iff L_{X_{T'}} is the pullback of an invertible sheaf on T'; load-bearing.

Source reference. Original lemma 0D23: sources/stacks-chow/tags.txt, source lines 16368–16368. Cited at proof lines 3317-3319: existence of Z in the invertible-module case; load-bearing.

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Lemma 0.2. *With assumptions and notation as in Proposition 0D2C. Then the diagonal* $\mathrm{Pic}_{X/B} \to \mathrm{Pic}_{X/B} \times_B \mathrm{Pic}_{X/B}$ *is representable by immersions. In other words,* $\mathrm{Pic}_{X/B} \to B$ *is locally separated.*

Proof. Let T be a scheme over B and let $s, t \in \mathrm{Pic}_{X/B}(T)$. We want to show that there exists a locally closed subscheme $Z \subset T$ such that $s|_Z = t|_Z$ and such that a morphism $T' \to T$ factors through Z if and only if $s|_{T'} = t|_{T'}$.

We first reduce the general problem to the case where s and t come from invertible modules on X_T. We suggest the reader skip this step. Choose an fppf covering $\{T_i \to T\}_{i \in I}$ such that $s|_{T_i}$ and $t|_{T_i}$ come from $\mathrm{Pic}(X_{T_i})$ for all i. Suppose that we can show the result for all the pairs $s|_{T_i}, t|_{T_i}$. Then we obtain locally closed subschemes $Z_i \subset T_i$ with the desired universal property. It follows that Z_i and Z_j have the same scheme theoretic inverse image in $T_i \times_T T_j$. This determines a descend datum on Z_i/T_i. Since $Z_i \to T_i$ is locally quasi-finite, it follows from More on Morphisms, Lemma 02W8 that we obtain a locally quasi-finite morphism $Z \to T$ recovering $Z_i \to T_i$ by base change. Then $Z \to T$ is an immersion by Descent, Lemma 02YM. Finally, because $\mathrm{Pic}_{X/B}$ is an fppf sheaf, we conclude that $s|_Z = t|_Z$ and that Z satisfies the universal property mentioned above.

Assume s and t come from invertible modules $\mathcal{V}$, $\mathcal{W}$ on X_T. Set $\mathcal{L} = \mathcal{V} \otimes \mathcal{W}^{\otimes -1}$ We are looking for a locally closed subscheme Z of T such that $T' \to T$ factors through Z if and only if $\mathcal{L}_{X_{T'}}$ is the pullback of an invertible sheaf on T', see Lemma 0D25. Hence the existence of Z follows from More on Morphisms of Spaces, Lemma 0D23. $\square$